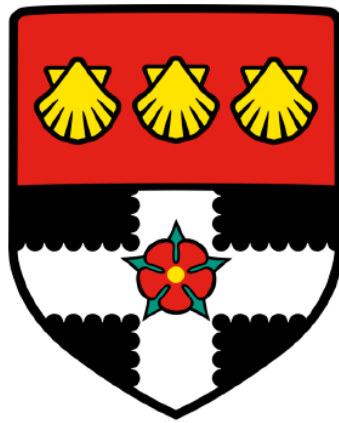


UNIVERSITY OF READING
DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

Owen Aljabar

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*A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Atmosphere, Ocean and Climate*

August 2024

Declaration of Authorship

I, Owen Aljabar, declare that this thesis titled, “Improving Surface Gravity Wave Forecasts Using Data Assimilation” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty Name

Department of Meteorology

Master of Science

Improving Surface Gravity Wave Forecasts Using Data Assimilation

by Owen Aljabar

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor. . .

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List of Abbreviations

DA	Data Assimilation
WW3	WAVEWATCH III
LETKF	Local Ensemble Transform Kalman Filter

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations form an important of many industries such as energy and research, and contribute to the economies of coastal countries round the world. Ocean waves pose a significant risk to offshore operations as they can lead to the loss of life and damage to equipment. This means that accurate wave predictions are a critical safety issue for offshore operations, the the operations can only be carried out safely when the waves are sufficiently calm. To provide these predictions the MetOcean Forecasting division of Fugro produces wave forecasts for a client undertaking offshore operations in the Gulf of Mexico. To avoid unnecessary downtime, they produce ensemble forecasts and apply corrections using a proprietary method based on the alpha factor method described in Wu and Gao, 2021.

This proprietary method is referred to by Fugro as a site-specific alpha factor and works by...

As well as these forecasts, Fugro has a buoy based near to the client which produces hourly observations of the sea state. Data assimilation refers to a range of techniques used to integrate observations into a forecast, with the aim of making the forecast more accurate. In this project, we explore the potential of DA to reduce uncertainty and error in these wave forecasts.

1.2 Previous work on DA for Wave Forecasts

Houghton et al., 2023 assimilated wave height observations from satellite altimeters and a global network of drifting Sofar spotter buoys. They compared two different DA methods, a deterministic optimal interpolation (OI) method (cite) and an ensemble-based local ensemble transform Kalman filter (LETKF) (cite). They found that both methods reduced forecast error out to 2.5 days leadtime, and the LETKF gave more physically realistic output states. In this project we will assimilate observations of both wave height and period in the hope that correlations between these variables might give better predictions of both. We will use the ensemble transform Kalman filter (ETKF).

Houghton et al., 2022 used OI to compare the effects of assimilating wave height and spectra. They found that both variables give very similar improvements in predictions of wave height, while spectral assimilation gives better results for other quantities such as period and direction. They also found that the spectral DA did particularly well in swell-dominated fields when compared to assimilating wave height. A plot showing the RMSE from both approaches is shown in figure 1.1

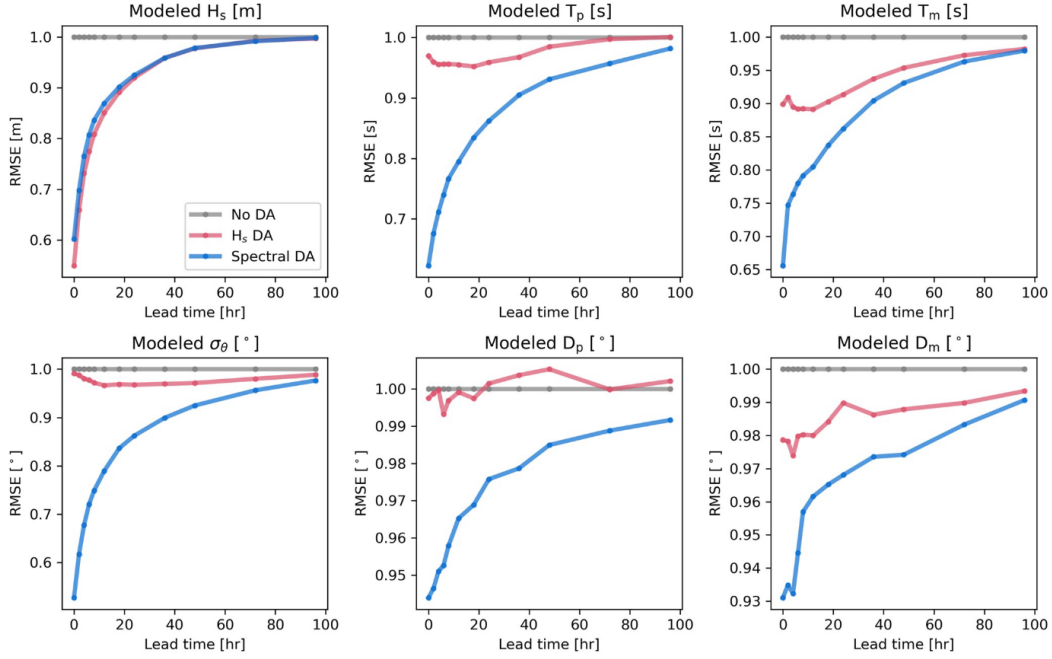


FIGURE 1.1: RMSE from spectral DA and wave height (H_s) DA, normalised by the RMSE without DA. Results shown for significant wave height H_s , peak period T_p , mean period T_m , directional spread σ_θ , peak direction D_p and mean direction D_m Figure 3 from Houghton et al. (2022)

Caires, Kim, and Groeneweg, 2018 investigated the effect on wave forecasts of using an ensemble Kalman filter (EnKF) method to adjust the wind fields used as forcing in a wave model. They found...

All the studies mentioned forecasted over a wide area (check). A problem with this approach is the inherent difficulty present in observing the ocean which means that observations are sparse, generally coming from drifting buoys which give irregular observations (reword) and satellite altimetry which can be unreliable (check). This is problematic for DA, which relies on regular and frequent observations to work well.

In this project we will be focusing mainly on a particular point of interest and analysing the effectiveness of DA to reduce forecast error at this point. The observations used in this project are also point-based, coming from a fixed buoy near the point of interest (how far?) which measures the sea state hourly. The point, which will be referred to as the buoy location, is of particular interest to a client of Fugro and is located near the Southern coast of the Gulf of Mexico. Its exact location cannot be disclosed due to confidentiality requirements. We hope that the regularity of observations and proximity of the buoy to the point of interest will lead to improved DA performance compared to area-based forecasts.

Another major difference between this study and others that should be noted is the difference in scale: all studies mentioned used many more observations than are used here, and

1.3 Main Research Questions

The main overarching research question in this project is “**What kind of reduction in forecast error can be achieved by assimilating Hs, Tm01 and Tm02 using the ensemble transform Kalman filter method?**”

To this end, we will use the ensemble transform Kalman filter (ETKF) to assimilate observations of the three variables and examine the errors resulting from this as compared to the errors before any DA is applied. Measures of error that will be examined are:

- The root mean square error (RMSE) of the ensemble mean. We expect the DA to move the ensemble mean closer to the true state so RMSE will be used to quantify and test this. The dependence of RMSE on forecast leadtime and location within the domain will be examined.
- The correlation between ensemble mean and the true state. DA should make the ensemble mean more closely correlated with the true state, so we will test this hypothesis. Correlations in space and in time will both be presented.
- Errors in maximum Hs. Extremes of Hs are of particular interest in these forecasts as these are the events which cause danger and require the pausing of offshore operations. We will examine the effectiveness of DA to reduce error in the timings and magnitudes of wave height maxima.

Chapter 2

Theory

This chapter outlines the theoretical aspects of this project, covering both the linear wave theory used in the wave model and the theory of the data assimilation methods used. The aim of the chapter is to describe all the theory required to understand the methods used in this project.

2.1 Ocean Waves

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. This section will describe some of the physical processes influencing ocean waves, as well as the theory of linear waves and wave energy spectra as modelled by the WAVEWATCH III (WW3) model used in this project.

2.1.1 Physical Processes

Ocean waves are an instance of surface gravity waves, meaning they propagate on the interface between two fluids of different densities. Their restoring force is gravity. The focus of this project is on waves with periods from 10s - 0.25s, which are generated by wind, although ocean waves can exist on a wide range of scales and be influenced by other factors such as surface tension and local air pressure variations (Holthuijsen, 2007, pp. 3-7).

There are multiple hypothesised physical mechanisms for the generation of waves by wind (e.g. ...). The approach from Miles is popular, and described in detail in Janssen (2004), chapter 3. Qualitatively speaking, Miles hypothesises that waves are generated by instability induced by the velocity shear between the air and ocean. Wind-generated waves will propagate in the same direction as the wind that generated them and will be of larger amplitude when the wind is stronger. (check)

The wind almost always acts as a source of wave energy, so there has to be a process present to dissipate waves and act as an energy sink. These processes are poorly understood but Janssen (2004), section 4.9 describes multiple physical mechanisms which may contribute to wave dissipation. One important process affecting dissipation is wave breaking, which occurs in shallow waters near shorelines. Water's molecular viscosity can lead to the damping of shorter waves and larger waves can be damped by the surface eddies induced by wave breaking.

Ocean waves in deep water are dispersive, and waves with longer wavelengths travel faster than waves with shorter wavelengths (equations 2.3-2.5). This leads to a phenomenon known as swell, which is the name given to long waves that have travelled

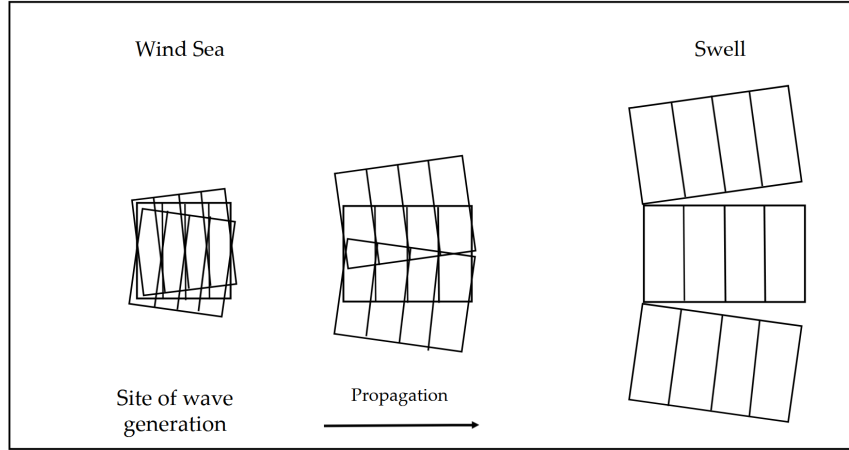


FIGURE 2.1: Schematic of the generation of wind and swell waves.

far from their source. Due to dispersion these waves have no short-wavelength component, so swell waves have longer wavelengths and periods than waves near the site of generation (known as wind sea). The other feature of swell waves is their regular appearance. While a wind sea has wave components in many different directions, by the time the waves have travelled away from the source these different components will spread out so that they do not overlap, and the resulting swell will be almost unidirectional. See figure 2.1 for a schematic.

2.1.2 Linear Wave Theory

This section will outline the linear wave theory relevant to this project, as derived in Vallis (2017), page 253. Linear wave theory is an approximation, and other higher-order theories of surface gravity waves exist, such as Stokes waves (cite) and cnoidal waves (cite). These higher-order theories will not be considered here since the WAVEWATCH III model used in this project models the ocean surface as a linear superposition of simple linear waves.

Parameters describing linear waves include the wavenumber k , direction θ and frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ i.e.

$$\mathbf{k} = k \cos \theta \hat{\mathbf{i}} + k \sin \theta \hat{\mathbf{j}} \quad (2.1)$$

where $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ are unit vectors pointing East and North respectively. This means that θ is the angle between $\mathbf{k}$ and $\hat{\mathbf{i}}$, measured anticlockwise from $\hat{\mathbf{i}}$. ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current, due to the Doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (2.2)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (2.3)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (2.4)$$

and the group speed may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (2.5)$$

The phase speed is the speed at which the wave peaks and troughs move, whereas the group speed is the speed at which wave groups move. The group speed is then the speed at which energy is transported by the wave. The wave travels in the direction of the wave vector $\mathbf{k}$ so the group and phase velocities are given by

$$\begin{aligned} \mathbf{c} &= c \cos \theta \hat{\mathbf{i}} + c \sin \theta \hat{\mathbf{j}} \\ \mathbf{c}_g &= c_g \cos \theta \hat{\mathbf{i}} + c_g \sin \theta \hat{\mathbf{j}} \end{aligned}$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.2 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

2.1.3 Wave Spectra

Note that equations 2.2 - 2.5 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the sea surface height variance associated with a particular frequency σ and direction θ , so the total height variance is

$$\overline{\eta^2} = \iint F d\sigma d\theta \quad (2.6)$$

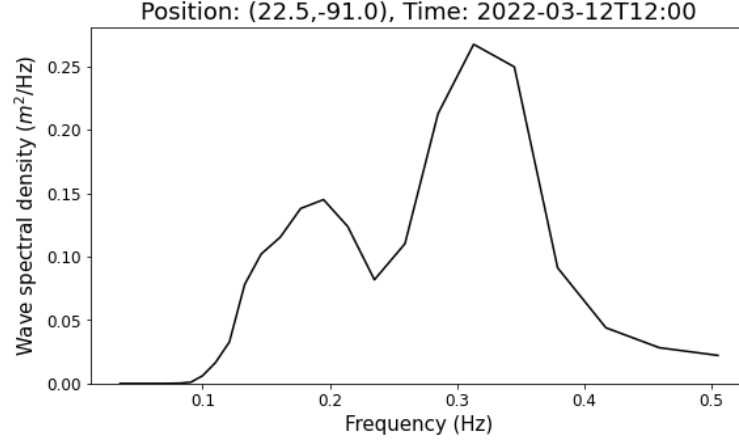


FIGURE 2.2: An example unidirectional spectrum from a point in the gulf of Mexico. Calculated from WW3 spectral forecast data.

This is related to the wave energy via $E = \rho g \overline{\eta^2}$. A rigorous definition of F is given in Holthuijsen (2007), section 3.5.6. One can also define the unidirectional spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$ which quantifies the surface height variance as a function of frequency. This can be integrated to calculate the energy contained in a specific frequency band $\Delta\sigma$:

$$\Delta E = \int_{\Delta\sigma} S(\sigma) d\sigma \quad (2.7)$$

S is a helpful diagnostic from which information about the appearance of the waves can be inferred. A widely spread spectrum implies irregular waves with many different frequency components whereas a tight spectrum implies regular waves with one dominant frequency. Low frequency waves correspond to swell, while high frequency waves correspond to a wind sea. An example of a unidirectional spectrum is plotted in figure 2.2.

Here, we also define the wave spectral moments which are derived from the unidirectional spectrum:

$$m_k = \int S(\sigma) \sigma^k d\sigma, m = \dots, -1, 0, 1, \dots \quad (2.8)$$

Note that $m_0 = \overline{\eta^2}$ is the overall surface height variance. These moments can be used to derive other important wave variables from the spectrum, such as significant wave height and wave period. In this project we focus on the significant wave height H_s and two measures of frequency Tm01, Tm02. These variables are defined below in equation 2.11 (expand?).

$$H_s = 4\sqrt{m_0} \quad (2.9)$$

$$Tm01 = \frac{m_0}{m_1} \quad (2.10)$$

$$Tm02 = \sqrt{\frac{m_0}{m_2}} \quad (2.11)$$

Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks (Bretherton and Garrett, 1968), so it is very useful for wave modelling. WW3 works by evolving the wave action according to the action of sources and sinks (see section 3.1).

2.2 Data Assimilation

As well as the model, we also have some direct observations of the waves which we would like to make use of to improve our predictions. We use techniques of data assimilation to achieve this.

We take a vector $\mathbf{x} \in \mathbb{R}^n$ to represent the state of the system, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$. We take a vector $\mathbf{y} \in \mathbb{R}^p$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^p$:

$$\mathbf{y} = \mathcal{H}(\mathbf{x}) + \varepsilon$$

where the error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. We call the vector spaces containing $\mathbf{x}$ and $\mathbf{y}$ the state space and observation space respectively. In operational contexts, the dimensions of these spaces are very large: typically $n \sim 10^8$ and $p \sim 10^6$. This means that considerable computational resources are required in most cases and computational efficiency is a key consideration in any operational DA implementation.

In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H} \in \mathbb{R}^{n \times p}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (2.12)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) \sim e^{-\frac{1}{2}(\|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (2.13)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (2.14)$$

The state that minimises this cost function is called the analysis $\mathbf{x}_a$ and may be expressed as

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{K}\mathbf{d}_b^o \quad (2.15)$$

where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain (derived in Asch, Bocquet, and Nodet, 2016, pp. 92-96). Note that the increment $\mathbf{d}_b^a = \mathbf{x}_a - \mathbf{x}_b$ is given by $\mathbf{B} \left(\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{d}_b^o \right)$, so it always lies in the column space of $\mathbf{B}$. This means that $\mathbf{B}$ is crucial to determining how information from the observations is spread around the domain by the DA system. In particular, if $\mathbf{B}$ is rank deficient, then all possible increments lie within a strict subspace of the state space. This reduces the degrees of freedom in the corrections the DA system can make, which limits the performance of the system.

DA methods which use equation 2.15 are called Kalman filters, in which context it is called the analysis step. Kalman filters generally alternate this analysis step with a forecast step in which a forecast model is used to evolve the state forward to the next observation time. In this project we do not have access to the forecast model, so we will only make use of the analysis step to update the state.

The specific method we will use is called the ensemble transform Kalman filter (ETKF) in which the analysis is performed in ensemble subspace rather than in the state space. Algebraically this means expressing the increment as a linear combination of ensemble perturbations

$$\mathbf{X}_f = \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_1^f - \bar{\mathbf{x}}^f, \mathbf{x}_2^f - \bar{\mathbf{x}}^f, \dots, \mathbf{x}_N^f - \bar{\mathbf{x}}^f \right) \quad (2.16)$$

$$\mathbf{x}^a = \bar{\mathbf{x}}^f + \mathbf{X}_f \mathbf{w}^a \quad (2.17)$$

Where $\mathbf{x}_i^f$ is the state of ensemble member i prior to assimilation and the ensemble mean $\bar{\mathbf{x}}^f$ is used as the background state. From Asch, Bocquet, and Nodet (2016) pp.163-164, the optimal coefficient vector $\mathbf{w}^a$ is given by

$$\mathbf{w}^a = \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-1} \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{d}_b^o \quad (2.18)$$

This method is algebraically equivalent to working in state space with the background error covariance matrix given by the forecast error covariance matrix

$$\mathbf{P}_f = \mathbf{X}_f \mathbf{X}_f^\top \quad (2.19)$$

This matrix is rank deficient since its column space has dimension equal to the ensemble size, but working in ensemble space has considerable advantages over working in state space in terms of computational cost (Hunt, Kostelich, and Szunyogh, 2007). The full analysis ensemble can then be generated from the analysis state derived in equation 2.18 via

$$\mathbf{x}_i^a = \mathbf{x}^a + \dots \quad (2.20)$$

And the analysis error covariance can $\mathbf{P}_a$ be defined similarly to $\mathbf{P}_f$ but from the analysis ensemble:

$$\mathbf{X}_a = \frac{1}{\sqrt{N-1}} (\mathbf{x}_1^a - \bar{\mathbf{x}}^a, \mathbf{x}_2^a - \bar{\mathbf{x}}^a, \dots, \mathbf{x}_N^a - \bar{\mathbf{x}}^a) \quad (2.21)$$

$$\mathbf{P}_a = \mathbf{X}_a \mathbf{X}_a^\top \quad (2.22)$$

Another advantage of the ETKF (compared to the stochastic ensemble Kalman filter) is that $\mathbf{P}_a$ matches exactly the form given by Kalman theory (cite?)

PROOF OF EQUIVALENCE WITH WEAK 4DENVAR

EnKF description - may not be needed

In most contexts, observations are not all simultaneous but instead come from a sequence of times $t = 0, 1, \dots$. One approach to assimilating non-simultaneous observations is to use a filter method such as the Ensemble Kalman Filter (Anderson, 2001; Evensen, 1994; Houtekamer and Mitchell, 1998). A filter method consists of successive analysis steps, in which the observations from time t are assimilated according to equation 2.15, alternating with update steps, in which the forecast model is used to evolve the state forward to time $t + 1$.

Where N is the ensemble size, $\mathbf{x}_t^{(i)}$ is the state of ensemble member i before the analysis step at time t and $\bar{\mathbf{x}}_t = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_t^{(i)}$ i.e. the ensemble mean.

In general, the observations from each time t may be available at different locations, or observations of different variables may be available, so the observation operator and observation error covariance matrix are functions of time. Denoting these at time t by $\mathbf{H}_t, \mathbf{R}_t$, the Kalman gain matrix takes the form

$$\mathbf{K}_t = \mathbf{P}_t^f \mathbf{H}_t^\top \left(\mathbf{H}_t \mathbf{P}_t^f \mathbf{H}_t^\top + \mathbf{R}_t \right)^{-1}$$

The EnKF method is prone to a problem called filter divergence - sampling error in the calculation of $\mathbf{P}_t^f$ means the magnitude of the forecast error covariance tends to erroneously shrink over time (Jazwinski, 1970). This means that after some time, the ensemble distribution becomes very tight and further observations do not affect the state of the ensemble, resulting in forecast analysis states which diverge from the true state. (schematic) To counteract this, we will employ the method of inflation described in Anderson and Anderson, 1999, where the forecast error covariance matrix is multiplied by a factor α before the analysis step is performed.

2.2.1 (Description of 4DEnVar - may not be needed)

In our case, we have observations from multiple times $t = 0, 1, \dots, T$ spread out over some time window, and we expect the state $\mathbf{x}$ to evolve over this time window. This evolution may be predicted using the model as follows: let $\mathbf{x}_0$ represent the state of the system at $t = 0$, then we define the model operator $\mathcal{M}_t$ for each t such that $\mathbf{x}_{t+1} = \mathcal{M}_t(\mathbf{x}_t)$ as predicted by the model. The cost function then becomes

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t\|_{\mathbf{R}_t^{-1}}^2 \quad (2.23)$$

$$= \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \mathbf{x}_0 \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.24)$$

where the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t , and $\hat{\mathbf{h}}_t = \mathbf{H}_t \mathcal{M}_{0 \rightarrow t}$ using the abbreviation $\mathcal{M}_{0 \rightarrow t} = \mathcal{M}_{t-1} \mathcal{M}_{t-2} \dots \mathcal{M}_0$.

We will use the 4DEnVar method, where a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned} \bar{\mathbf{x}}_b &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_b^{(n)} \\ \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_b^{(1)} - \bar{\mathbf{x}}_b, \mathbf{x}_b^{(2)} - \bar{\mathbf{x}}_b, \dots, \mathbf{x}_b^{(N)} - \bar{\mathbf{x}}_b \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \chi \end{aligned}$$

Here $\mathbf{x}_b^{(n)}$ represents the background state from ensemble member n , and N is the ensemble size. The cost function can then be rewritten in terms of χ and simplified by linearising $\hat{\mathbf{h}}_t$ around $\bar{\mathbf{x}}_b$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \bar{\mathbf{x}}_b - \hat{\mathbf{H}}_t \mathbf{X}'_b \chi \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.25)$$

where $\hat{\mathbf{H}}_t = \mathbf{H}_t \mathbf{M}_{0 \rightarrow t}$ is the linearisation of $\hat{\mathbf{h}}_t$. This cost function is simplified by taking a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}'_b \mathbf{X}_b'^\top$, which reduces to the identity under the control variable transform. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right) \approx \mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} \right) - \overline{\mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b \right)}$$

Since $\mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right)$ is the n^{th} column of $\hat{\mathbf{H}}_t \mathbf{X}'_b$, this approximation allows us to compute the cost function and its gradient without a tangent linear or adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes (Bannister, 2017).

Chapter 3

Experimental Setup

This chapter is intended to describe the details of the experimental setup used in this project. It contains a description of the wave model used to generate the forecasts, as well as an overview of the background state, the observations used and the errors present in these observations. Finally a description is given of the twin experiment method used in this project, as well as the data assimilation setup.

3.1 Model Description and Domain

Waves are modelled by Fugro using WW3 (WW3DG, 2019), which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described in section 2.1.3. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - Janssen, 2004)

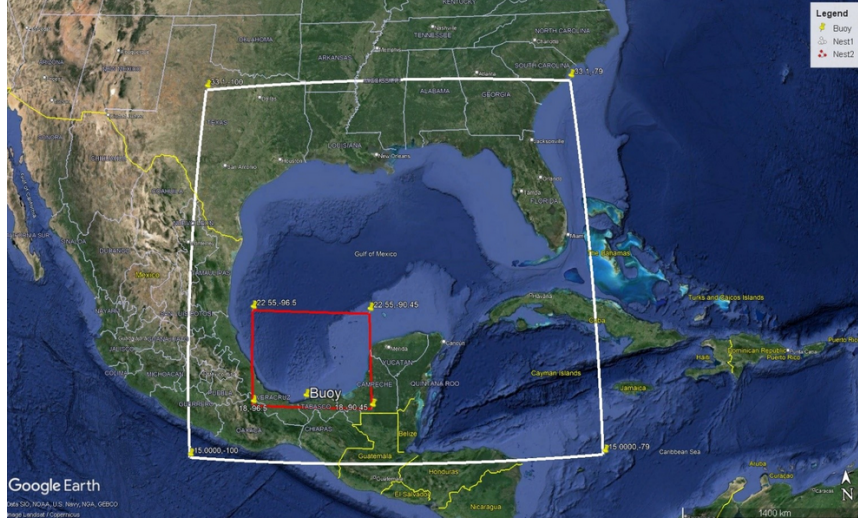


FIGURE 3.1: Model domains shown on map.

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables we will be focusing on in this study are significant wave height H_s , and two measures of the dominant wave period T_{m01} and T_{m02} . These are related to the spectrum via the spectral moments (equations 2.8, 2.11).

The reason for assimilating both T_{m01} and T_{m02} is that they are defined from different spectral moments, so assimilating both will provide more information about the spectrum than either one alone, which should lead to better DA results.

As forcing, WW3 uses just the 10 metre wind, and this comes from the GEFS ensemble. The GEFS forecast contains one control run plus 20 perturbed ensemble members. Each GEFS run acts as forcing for one member of the wave ensemble. (Do they?)

The domain in which forecasts are produced lies between $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$ in the gulf of Mexico. Fugro use a nested domain to produce the forecasts, running WW3 on a 0.5° resolution globally rectangular grid, and two higher resolution nested grids covering the region of interest. There is a 0.1° resolution grid spanning the region described above, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. We have seven of these daily runs available, with start dates spanning from 06/03/2022 to 12/03/2022. (include schematic) As we do not have access to the model code in this project, we will treat each of these runs as one 'timestep' of the DA system. Further description of how this works can be found in section 3.5.

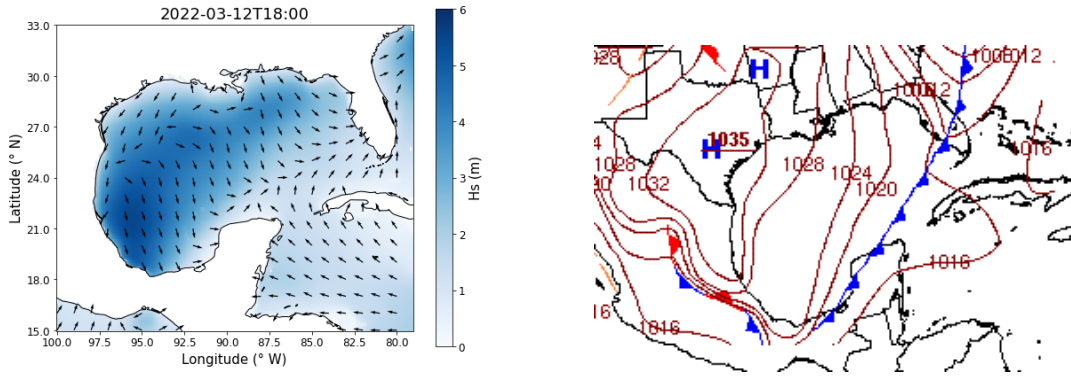


FIGURE 3.2: Left: Hs (colours) and direction (arrows) in the GoM at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive

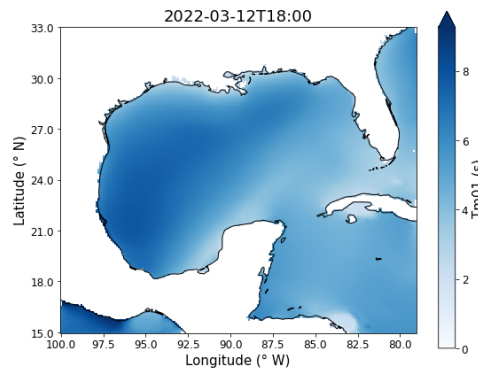


FIGURE 3.3: A map of the Gulf of Mexico with Tm01 at 18:00 on 06/03/2022 shown with colours.

3.2 Background State and Climatology

In this section we present a general overview of the background state and climatology in the domain. This will mostly come from the model data which spans a period from 06/03/2022 to 19/03/2022. Since the available observations span a longer period of roughly six months, observations will also be used to identify broader climatological features at the buoy location.

3.2.1 Model Background State

The model output has largely westerly waves throughout much of the period (check, other climatologies).

There is a large peak in Hs on the 12/03/2022, related to a cold front passing through the region. A map of Hs in the GoM at 18:00 on 06/03/2022 is shown in figure 3.2, along with a surface chart from the same time. The cold front is visible on the chart, and high waves are present in a large region behind the front. This is in line with Appendini et al. (2014), a study into the wave climatology of the GoM. In that paper, cold fronts are identified as a common driver of high waves between October and April, with the most intense peaks occurring between December and March.

The high waves associated with this cold front have a long period of ~ 8 s, which suggests they are swell waves. This is illustrated in figure 3.3

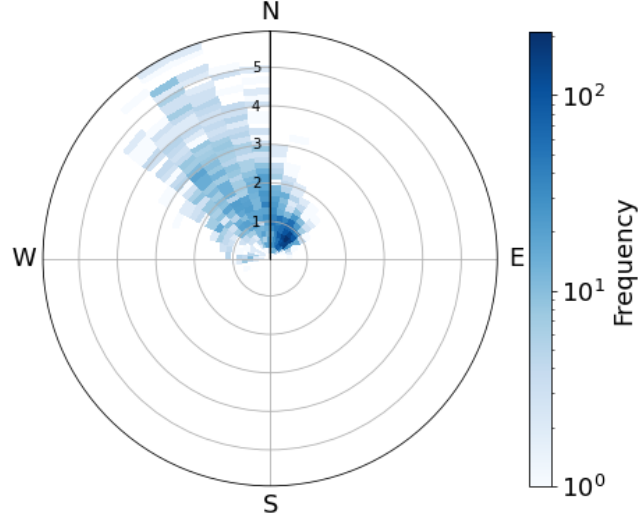


FIGURE 3.4: Density plot of H_s (radial axis) and wave direction at the buoy location.

3.2.2 Climatology at Buoy Location

Figure 3.4 shows a density plot of observed H_s and wave direction at the buoy location. There is a clear peak in frequency corresponding to waves of height $\sim 0.5\text{m}$ coming from a northeasterly direction. Some higher waves (up to $\sim 5\text{m}$) are present coming from a north-northwesterly direction.

3.3 ADCP Observations

Our observations come from an acoustic doppler current profiler (ADCP) buoy, which estimates the spectrum by measuring the wave induced velocity and surface height variations. According to Vallis, 2017, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 g \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 2.6 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

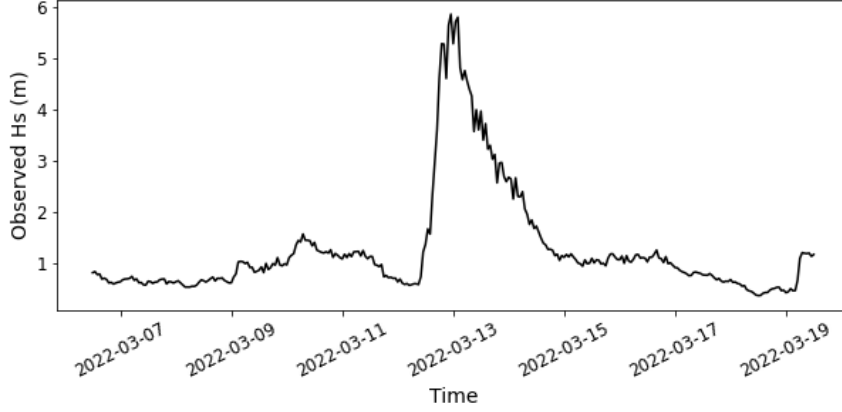


FIGURE 3.5: Observed timeseries of significant wave height from ADCP buoy

Other variables are then deduced from this spectrum, including H_s , T_{m01} and T_{m02} as described above.

We have observations available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.5. The peak in wave height late on the 12th of March is clearly visible on the plot.

3.3.1 Diagnosis of Observation Errors

To simplify the computation, we assume that the observation error covariance matrix $\mathbf{R}$ is diagonal, essentially ignoring covariances between observations taken at different times. This also makes determining $\mathbf{R}$ easier since all that is required is to estimate the observation error variance at each point. To determine the magnitude of the observation errors, several methods are available. A simple one is to plot a histogram of observation - ensemble mean, called an $o - b$ plot. The spread in $o - b$ comes from a combination of observation and background error, so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this error, say 50%. An $o - b$ histogram for significant wave height is shown in figure 3.6. From the plot we see that the spread on the histogram is approximately 1m, so the standard of observational errors in significant wave height may be estimated to be approximately 50cm.

This approach may be quantified more precisely following Desroziers et al., 2006, by defining $d_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ and using the relation

$$\mathbb{E} \left[d_b^o (d_b^o)^\top \right] = \mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R} \quad (3.2)$$

Taking traces and dividing by the number of observations, we get a good estimate for the observation error variance which is

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} - \frac{\text{Tr} (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top)}{\# \text{ observations}} \quad (3.3)$$

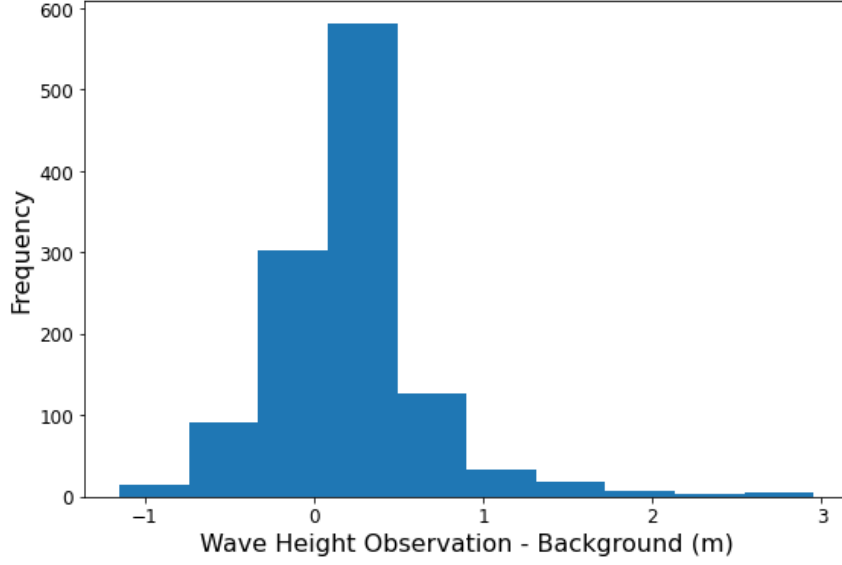


FIGURE 3.6: Observation - background histogram for significant wave height.

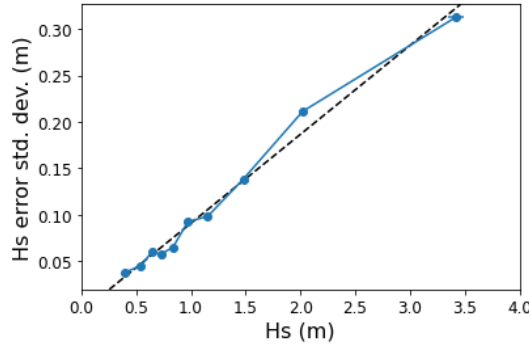


FIGURE 3.7: A plot of H_s error standard deviation against H_s .

In our case, the observation operator is very simple and simply selects the forecast values from the appropriate location and times, so the diagonal elements of $HP^f H^\top$ are simply the ensemble forecast variances at the observation location and times.

Another available method is that used in Houghton et al., 2023. They used a collocation study to estimate the error in wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average wave height of each pair. The difference of each pair was taken, and the standard deviation of differences from each bin was computed. They found a strong dependence of error on wave height.

In our case, we only have observations from one buoy and they are available hourly, so we take the pairs to be observations one hour apart. We then calculate the standard deviation using the method described above to obtain the plot in figure 3.7. As in Houghton et al., 2023, we find a strong linear relationship between errors and wave height, with the error standard deviation being approximately 9.6% of the wave height. Repeating this process with the periods T_{m01} and T_{m02} yields the plots shown in figure 3.8.

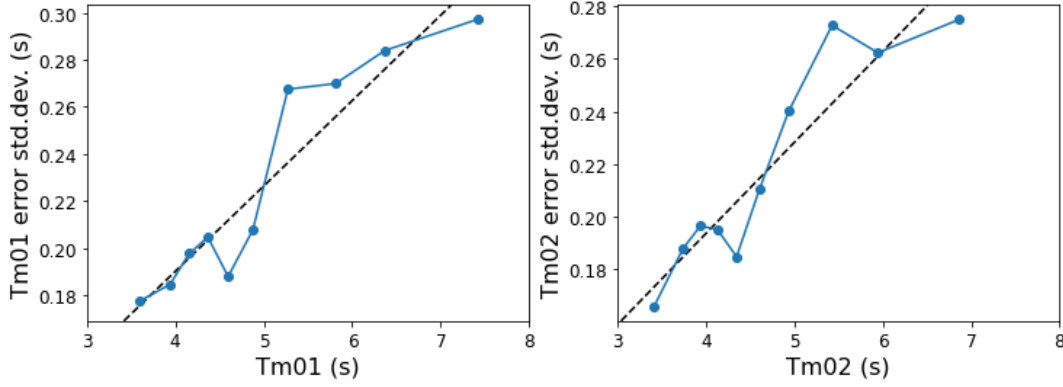


FIGURE 3.8: Plots of Tm01 and Tm02 error standard deviation against Tm01 and Tm02 respectively.

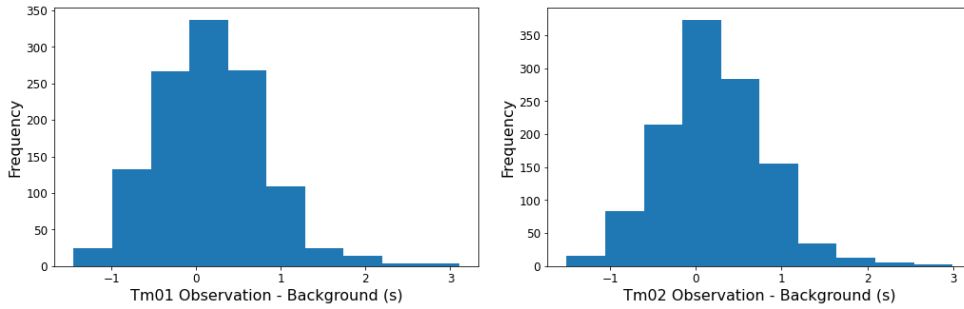


FIGURE 3.9: Observation minus background histograms for Tm01 (left) and Tm02 (right).

The error distributions for Tm01 and Tm02 also fall approximately on straight lines, though the fit is not as good as for Hs. The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both. So we will model the error standard deviation in Hs as being equal to 10% of Hs, and the error standard deviation in Tm01 and Tm02 as 3.5% of Tm01 and Tm02 respectively.

Since the errors in period are such a small percentage of the observations, it is possible that at low periods the error may become very small, which would lead to the DA system fitting the observations too closely and producing physically unrealistic increments. To avoid this, we set a minimum error threshold ε_0 and set the observation error standard deviation equal to this threshold whenever the scaling method gives a value less than ε_0 . This threshold is determined using the method from Desroziers et al., 2006, using only the observations of wave height with a value of less than 5 seconds, as these are the observations which may lead to small errors.

The terms in 3.3 were calculated to be

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right]}{\# \text{ observations}} = 0.16s^2, \quad \frac{\text{Tr}(\mathbf{H}\mathbf{P}^f\mathbf{H}^\top)}{\# \text{ observations}} = 0.11s^2$$

So the observation error variance is estimated to be $0.05s^2$ and the standard deviation threshold is set at $0.23s$.

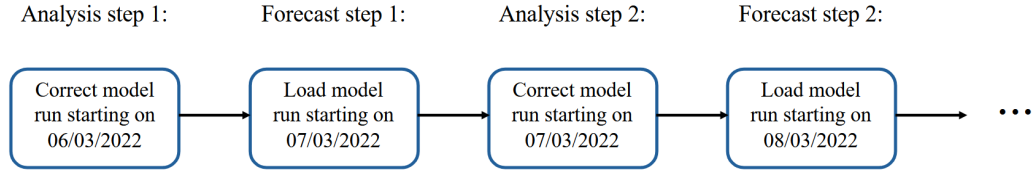


FIGURE 3.10: Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.

3.4 Twin Experiment

In this study, we will not be using real observations in our assimilation but setting aside one ensemble member to represent the ‘truth’ and generating fictional observations based on this forecast data. This is known as a twin experiment, and has several advantages over assimilating real observations:

- The ‘truth’ is available away from the location of observations, so forecast error can be calculated over the whole model domain. This allows the effectiveness of DA to improve forecasts away from observation locations to be evaluated

The observations are generated by sampling values from the truth at predefined observation locations and times. Here we sample hourly values from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from a normal distribution with standard deviation equal to the error standard deviation derived in section 3.3

3.5 Data Assimilation Setup

To perform the data assimilation we use pyPDAF (Chen, Nerger, and Lawless, 2024), which is a python interface for the fortran-based Parallel Data Assimilation Framework (PDAF). We will use the ETKF method which comes builtin with PDAF. As we are not able to run the WAVEWATCH III model ourselves, at each analysis step we will correct one 7-day forecast window all at once, and then the forecast step will consist of selecting data from the next 7-day window. In this way the time within the 7-day window is treated like another spatial coordinate by the DA system. A schematic of this process is included in figure 3.10

At each analysis step, we will assimilate observations from the first three days of the forecast window. This allows us to investigate errors at times after the assimilation window and assess the capability of DA to improve the forecast at future time when observations are unavailable.

Chapter 4

Results

This chapter will contain an analysis of the errors in the DA output as compared to the model output with no DA.

4.1 RMSE

4.2 Correlation

4.3 Errors in Maximum Hs

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

```
\hypersetup{urlcolor=red}, or
```

```
\hypersetup{citecolor=green}, or
```

```
\hypersetup{allcolor=blue}.
```

If you want to completely hide the links, you can use:

```
\hypersetup{allcolors=}, or even better:
```

```
\hypersetup{hidelinks}.
```

If you want to have obvious links in the PDF but not the printed text, use:

```
\hypersetup{colorlinks=false}.
```


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